Natalyx MVP Fixes

Work Package 08: Payments, invoices, credits, and refunds

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2026-07-12

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1 Work Package 08: Payments, invoices, credits, and refunds

Review draft only. Ownership and sequencing remain recommendations until Luke and Alan approve the plan together.

Plan	Value
Work package	WP08
Plan file	`WP08 - Payments, invoices, credits, and refunds.md`
Proposed PRs	P08-01, P08-02, P08-03, P08-04
Inspection snapshot	2026-07-12 19:30 America/New_York

1.1 Purpose and outcome

Problem. No append-only commercial ledger or processor-neutral payment boundary currently supports the $8,900 JMF, $750 enrollment, Node 0 credit, milestones, deterministic refunds, or failure recovery.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. Immutable commercial ledgers, idempotent provider inbox, milestone engine, reproducible credits/refunds, reconciliation, and safe non-critical pause behavior.

Explicit non-goals.

- No custody of pass-through funds

- No financing assumption
- No pregnancy-loss automation before policy
- No accounting-general-ledger replacement

1.2 Verified current-state evidence

Classification is explicit: **verified** is observed code/PR/live behavior; **verified limitation** records an evidence gap; **design authority** is an approved planning source; **design decision** is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-025	design authority	business model / Natalyx_Payment_and_Business_Model_BYOS_MVP.docx v1.0	Sections 3-12 and implementation tables	Node 0 Free must complete the match; Complete is optional/creditable; $8,900 JMF and $750 enrollment anchor the configurable model; promotion follows explicit commitment and funding, not willingness or redirect.

1.3 Architecture and contracts

- Payments, credits, refunds, adjustments, invoices, and commercial events are append-only; mutable balances are derived projections.
- Processor events enter an idempotent inbox and become internal commands/events only after signature and amount/currency checks.
- Billing follows durable case evidence and never acts as first notification of clinical events.

Contract area	Package rule
Ownership boundary	`app`; repository-specific PRs remain separate.
Entities/state	`journey_commitment`, `payment_obligation` + `append-only entries`, `workspace_entitlement`
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Processor webhook and internal payment commands. Internal evidence-to-invoice commands and explainable refund quote.
Storage	Append-only financial ledgers and unique idempotency constraints. Processor-event inbox/outbox and mapping records.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.

Contract area	Package rule
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

1.3.1 Applicable state machines

1.3.2 Entitlement (`workspace_entitlement`)

Owner: **natalyx-app commercial domain**. States: `free`, `complete_pending`, `complete_active`, `suspended`, `revoked`.

Transition: `none` -> `free`

- **Actor:** system
- **Evidence:** active Node 0 workspace
- **Idempotency key:** `entitlement:{workspace}:free`
- **Invalid transitions:** No workspace
- **Audit event:** `entitlement_granted`
- **Retry:** Return current
- **Recovery:** Repair projection from event

Transition: `free` -> `complete_pending`

- **Actor:** IP/commercial service
- **Evidence:** accepted Complete offer/config
- **Idempotency key:** `entitlement:{workspace}:{commitment}`
- **Invalid transitions:** Unknown price/config/terms
- **Audit event:** `complete_requested`
- **Retry:** Return pending
- **Recovery:** Cancel/expire commitment

Transition: `complete_pending` -> `complete_active`

- **Actor:** payment domain
- **Evidence:** settled Complete ledger entry
- **Idempotency key:** `entitlement:{payment}:activate`
- **Invalid transitions:** Amount/currency/config mismatch
- **Audit event:** `complete_activated`
- **Retry:** Replay no-op
- **Recovery:** Reconcile and retry from inbox

Transition: `complete_active` -> `suspended`

- **Actor:** authorized commercial policy
- **Evidence:** refund/reversal or approved pause reason
- **Idempotency key:** `entitlement:{id}:suspend:{event}`
- **Invalid transitions:** Cannot remove free/safety capability
- **Audit event:** `entitlement_suspended`
- **Retry:** No-op if same event
- **Recovery:** Reinstate or revoke through new event

Transition: `complete_active|suspended -> revoked`

- **Actor:** authorized operator/policy
- **Evidence:** fraud/refund/terms revocation evidence
- **Idempotency key:** `entitlement:{id}:revoke:{event}`
- **Invalid transitions:** No silent case cancellation
- **Audit event:** `entitlement_revoked`
- **Retry:** Idempotent
- **Recovery:** Manual reviewed regrant creates new entitlement event

1.3.3 Commercial commitment (`journey_commitment`)

Owner: **natalyx-app commercial domain**. States: `draft`, `accepted`, `funding_pending`, `funded`, `cancelled`, `expired`.

Transition: `none -> draft`

- **Actor:** IP
- **Evidence:** active workspace + displayed config/terms
- **Idempotency key:** `commitment:{workspace}:{config}`
- **Invalid transitions:** No active config/workspace
- **Audit event:** `commitment_drafted`
- **Retry:** Return existing draft
- **Recovery:** Create new draft after expiry/version change

Transition: `draft -> accepted`

- **Actor:** IP
- **Evidence:** terms/privacy consent + payment path
- **Idempotency key:** `commitment:{id}:accept:{terms_hash}`
- **Invalid transitions:** Stale/missing acceptance
- **Audit event:** `commitment_accepted`
- **Retry:** Return accepted
- **Recovery:** New acceptance revision if version changes

Transition: `accepted -> funding_pending`

- **Actor:** commercial service
- **Evidence:** enrollment invoice/payment intent created

- **Idempotency key:** `commitment:{id}:funding`
- **Invalid transitions:** No invoice or invalid amount
- **Audit event:** `funding_requested`
- **Retry:** Resume same invoice
- **Recovery:** Cancel/reissue under audited adjustment

Transition: `funding_pending` -> `funded`

- **Actor:** payment domain
- **Evidence:** net settled enrollment funding meets versioned requirement
- **Idempotency key:** `enrollment-funding:{id}:v1`
- **Invalid transitions:** Redirect/pending/failed/reversed evidence
- **Audit event:** `commercial.enrollment_funding_confirmed.v1`
- **Retry:** Exactly-once event/outbox replay
- **Recovery:** Reconcile inbox/ledger; no browser repair

Transition: `draft|accepted|funding_pending` -> `cancelled`

- **Actor:** IP/authorized operator
- **Evidence:** explicit cancellation and refund policy
- **Idempotency key:** `commitment:{id}:cancel:{request}`
- **Invalid transitions:** Already funded/promoted uses case cancellation rail
- **Audit event:** `commitment_cancelled`
- **Retry:** Idempotent
- **Recovery:** New commitment required

Transition: `draft|accepted` -> `expired`

- **Actor:** system
- **Evidence:** offer/terms TTL or config deactivation
- **Idempotency key:** `commitment:{id}:expire`
- **Invalid transitions:** No expiry after funding
- **Audit event:** `commitment_expired`
- **Retry:** Idempotent
- **Recovery:** Draft new version

1.3.4 Payment (`payment_obligation + append-only entries`)

Owner: **natalyx-app payment domain**. States: `created`, `processing`, `settled`, `failed`, `partially_refunded`, `refunded`, `reversed`.

Transition: `none` -> `created`

- **Actor:** billing policy
- **Evidence:** valid invoice/obligation
- **Idempotency key:** `payment:{invoice}:create`
- **Invalid transitions:** Duplicate or invalid invoice

- **Audit event:** `payment_created`
- **Retry:** Return current intent
- **Recovery:** Void/reissue through adjustment

Transition: `created|failed -> processing`

- **Actor:** processor adapter
- **Evidence:** signed provider acknowledgement
- **Idempotency key:** `processor:{provider_event}`
- **Invalid transitions:** Amount/currency/customer mismatch
- **Audit event:** `payment_processing`
- **Retry:** Inbox replay no-op
- **Recovery:** Retry same intent or audited replacement

Transition: `processing|created -> settled`

- **Actor:** payment domain
- **Evidence:** verified settlement event
- **Idempotency key:** `processor:{provider_event}`
- **Invalid transitions:** Browser redirect or unverified event
- **Audit event:** `payment_settled`
- **Retry:** Exactly-once append
- **Recovery:** Reconcile provider and append repair event

Transition: `created|processing -> failed`

- **Actor:** payment domain
- **Evidence:** verified failure/timeout policy
- **Idempotency key:** `processor:{provider_event}`
- **Invalid transitions:** Cannot overwrite settled
- **Audit event:** `payment_failed`
- **Retry:** Replay no-op
- **Recovery:** Retry/new attempt linked to obligation

Transition: `settled|partially_refunded -> partially_refunded|refunded`

- **Actor:** authorized refund service
- **Evidence:** approved refund calculation + processor confirmation
- **Idempotency key:** `refund:{refund_request}`
- **Invalid transitions:** Exceeds settled less prior refunds
- **Audit event:** `refund_settled`
- **Retry:** Replay no-op
- **Recovery:** Reconcile/adjust; never mutate original

Transition: `settled -> reversed`

- **Actor:** payment domain

- **Evidence:** verified chargeback/reversal
- **Idempotency key:** `processor:{provider_event}`
- **Invalid transitions:** Unsupported/manual assertion
- **Audit event:** `payment_reversed`
- **Retry:** Replay no-op
- **Recovery:** Open discrepancy and apply policy; case is not deleted

1.3.5 Append-only commercial contract

Authoritative ledgers:

- `commercial_events`
- `processor_events`
- `invoice_events`
- `payment_ledger_entries`
- `credit_ledger_entries`
- `refund_ledger_entries`
- `adjustment_ledger_entries`
- `performance_obligation_events`
- `promotion_events`

Derived projections:

- payment plan state
- invoice status
- settled cash
- available credit
- recognized/deferred amount
- refundable amount
- delinquency/pause status

Rules:

- Every record carries currency in integer minor units, commercial configuration version, source/idempotency/correlation identifiers and immutable timestamps.
- Mutable balances/statuses are projections only; reversals, chargebacks, credits, refunds and adjustments append entries.
- Processor events are signature-verified, stored once and normalized before domain use.
- Node 0 Complete credit cannot create a negative JMF obligation and stays distinct from cash/refund entries.
- Pass-through clinic/legal/psych/background/escrow/insurance/GC funds never enter Natalyx revenue ledgers.
- Clinical-linked invoices require durable evidence plus dignity-of-timing delay and can never be the first clinical notification.
- Refundable cash = max(0, settled JMF cash - earned JMF - prior cash refunds - chargebacks/reversals), with policy hooks for unresolved cases.

1.4 Explicit PR sequence

Order	ID	Title	Repository	Dependencies	Gate	Local preparation	Merge readiness
1	P08-01	Add append-only commercial and money ledgers	app	P02-03, P07-01	A2	Not ready; named PR, decision or programme-acceptance prerequisites remain	Blocked pending PR dependencies P02-03, P07-01; Gate A0 active and verified; Alan review
2	P08-02	Add processor-neutral payment command and webhook inbox	app	P08-01, P07-02	A2	Not ready; named PR, decision or programme-acceptance prerequisites remain	Blocked pending PR dependencies P08-01, P07-02; technical decisions/controls D-PAY-01; Gate A0 active and verified; Alan review
3	P08-03	Implement milestone invoicing, credits, and deterministic refunds	app	P08-02, P07-04, P09-04	A2	Not ready; named PR, decision or programme-acceptance prerequisites remain	Blocked pending PR dependencies P08-02, P07-04, P09-04; Gate A0 active and verified; Alan review
4	P08-04	Add payment reconciliation and safe service-pause operations	app	P08-03	A2	Not ready; named PR, decision or programme-acceptance prerequisites remain	Blocked pending PR dependencies P08-03; Gate A0 active and verified; Alan review

1.4.1 P08-01 - Add append-only commercial and money ledgers

Field	Plan
Repository / base	`app` (`main`)
Gate / size	A2 / Exceptional
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-03, P07-01; Gate A0 active and verified; Alan review
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P02-03, P07-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Payment/business-model DOCX

1.4.1.1 Objective Create immutable commercial-event, invoice, payment, credit, refund, and adjustment entries with deterministic projections and config-version references.

1.4.1.2 Scope

- Ledger invariants and currency precision
- Derived balances/status projections
- Audit/source/idempotency references

1.4.1.3 Explicit non-goals

- No processor
- No mutable balance authority
- No pass-through custody

1.4.1.4 Expected files or modules

- `migrations/versions/*_commercial_ledgers.py`
- `app/storage/ledgers.py`
- `app/domain/money.py`
- `tests/python/test_ledgers.py`

1.4.1.5 Change map

Area	Planned change
Schema/model	Append-only financial ledgers and unique idempotency constraints.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Apply through the named versioned migration; no ad-hoc startup DDL. Backfill is none unless explicitly listed in scope.
Security/privacy	Least-privilege write paths and redacted logs.

Area	Planned change
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

1.4.1.6 Tests and validation

Objective tests:

- Double-entry-like conservation checks
- duplicate key
- credit floor
- refund totals
- concurrent append

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate **A2** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

1.4.1.7 Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

1.4.1.8 Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

1.4.1.9 Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

1.4.1.10 Dependencies and parallel work

Hard dependencies: P02-03, P07-01. Parallel opportunities: P08-03, P08-04

1.4.1.11 Objective acceptance criteria

- ☐ Create immutable commercial-event, invoice, payment, credit, refund, and adjustment entries with deterministic projections and config-version references.
- ☐ Every listed objective test passes locally and at the required gate where applicable.
- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- ☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

1.4.1.12 Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	payments, financial ledgers, PostgreSQL
Higher-capability review	Mandatory before merge

1.4.2 P08-02 - Add processor-neutral payment command and webhook inbox

Field	Plan
Repository / base	`app` (`main`)
Gate / size	A2 / Exceptional
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P08-01, P07-02; technical decisions/controls D-PAY-01; Gate A0 active and verified; Alan review
Technical-decision owner	D-PAY-01: Alan - later processor, security and settlement mechanism
Dependencies	P08-01, P07-02
Blocking decisions/checkpoints	D-PAY-01
Source / salvage	Business-model processor-neutral requirement

1.4.2.1 Objective Define provider ports and a signed, idempotent webhook inbox that emits internal settlement/failure/refund events, including commercial.enrollment_funding_confirmed.v1 when domain conditions hold.

1.4.2.2 Scope

- Provider-neutral intents and event mapping
- Signature/replay/order handling
- Settlement predicate and outbox

1.4.2.3 Explicit non-goals

- No browser redirect settlement
- No financing implementation unless separately approved

1.4.2.4 Expected files or modules

- `app/payments/ports.py`
- `app/payments/webhook_inbox.py`
- `app/routes/payment_webhooks.py`
- `tests/python/test_payment_webhooks.py`

1.4.2.5 Change map

Area	Planned change
Schema/model	Processor-event inbox/outbox and mapping records.
API/routes	Processor webhook and internal payment commands.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	Secrets/endpoints deferred to guarded environment configuration.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

1.4.2.6 Tests and validation Objective tests:

- Signature failure
- replay/out-of-order
- amount/currency mismatch
- timeout/retry
- settlement event exactly once

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate **A2** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

1.4.2.7 Diagnostics and observability Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

1.4.2.8 Rollout Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

1.4.2.9 Rollback or forward repair Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

1.4.2.10 Dependencies and parallel work

Hard dependencies: P08-01, P07-02. Parallel opportunities: P08-04

1.4.2.11 Objective acceptance criteria

- ☐ Define provider ports and a signed, idempotent webhook inbox that emits internal settlement/failure/refund events, including commercial.enrollment_funding_confirmed.v1 when domain conditions hold.
- ☐ Every listed objective test passes locally and at the required gate where applicable.
- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- ☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

1.4.2.12 Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	D-PAY-01: Alan - later processor, security and settlement mechanism
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	payment security, webhooks, event systems
Higher-capability review	Mandatory before merge

1.4.3 P08-03 - Implement milestone invoicing, credits, and deterministic refunds

Field	Plan
Repository / base	`app` (`main`)
Gate / size	A2 / Exceptional
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P08-02, P07-04, P09-04; Gate A0 active and verified; Alan review
Technical-decision owner	D-PAY-01: Alan - later processor, security and settlement mechanism
Dependencies	P08-02, P07-04, P09-04
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Payment/business-model milestone/refund tables

1.4.3.1 Objective

Turn explicit evidence-backed case transitions into one invoice each, apply Node 0 credit, and calculate reproducible earned/uncharged/refunded amounts from immutable state.

1.4.3.2 Scope

- Enrollment/activation/clearance/legal-escrow/pregnancy/birth milestones
- Credit allocation
- Non-clearance and cancellation tables
- Dignity delay for clinical-linked invoices

1.4.3.3 Explicit non-goals

- No unresolved pregnancy-loss automatic disposition
- No promise of reimbursement

1.4.3.4 Expected files or modules

- `app/domain/billing_policy.py`
- `app/domain/refunds.py`
- `app/event_handlers/case_billing.py`
- `tests/python/test_billing_policy.py`

1.4.3.5 Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Internal evidence-to-invoice commands and explainable refund quote.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Only authorized domain events trigger billing.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

1.4.3.6 Tests and validation

Objective tests:

- All milestone/refund rows
- duplicate/stale transition
- Node 0 credit
- pay-in-full/deposit math
- clinical notification separation

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate **A2** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

1.4.3.7 Diagnostics and observability Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

1.4.3.8 Rollout Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

1.4.3.9 Rollback or forward repair Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

1.4.3.10 Dependencies and parallel work Hard dependencies: P08-02, P07-04, P09-04. Parallel opportunities: P08-01

1.4.3.11 Objective acceptance criteria

- ☐ Turn explicit evidence-backed case transitions into one invoice each, apply Node 0 credit, and calculate reproducible earned/uncharged/refunded amounts from immutable state.
- ☐ Every listed objective test passes locally and at the required gate where applicable.
- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- ☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

1.4.3.12 Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	D-PAY-01: Alan - later processor, security and settlement mechanism
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	payments, commercial policy, case state
Higher-capability review	Mandatory before merge

1.4.4 P08-04 - Add payment reconciliation and safe service-pause operations

Field	Plan
Repository / base	`app` (`main`)
Gate / size	A2 / Large

Field	Plan
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P08-03; Gate A0 active and verified; Alan review
Technical-decision owner	D-PAY-01: Alan - later processor, security and settlement mechanism
Dependencies	P08-03
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Business-model failure handling

1.4.4.1 Objective Expose operator-safe reconciliation, discrepancy, receipt, retry, and non-critical pause controls without blocking safety information or altering clinical truth.

1.4.4.2 Scope

- Ledger-vs-processor reconciliation
- Itemized receipt data
- Grace/reminder/soft-pause policy
- Audited manual adjustments

1.4.4.3 Explicit non-goals

- No organization-wide ambient access
- No critical-service lockout

1.4.4.4 Expected files or modules

- `app/payments/reconciliation.py`
- `app/routes/operator_billing.py`
- `app/templates/internal/billing/`
- `tests/python/test_reconciliation.py`

1.4.4.5 Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	Internal role-gated discrepancy and adjustment surface.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

1.4.4.6 Tests and validation

Objective tests:

- Role denial
- mismatch workflow
- manual adjustment audit
- critical content remains accessible
- provider outage

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate **A2** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

1.4.4.7 Diagnostics and observability

Metrics/alerts for settlement, refund, mismatch and reserve inputs.

1.4.4.8 Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

1.4.4.9 Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

1.4.4.10 Dependencies and parallel work

Hard dependencies: P08-03. Parallel opportunities: P08-01, P08-02

1.4.4.11 Objective acceptance criteria

- ☐ Expose operator-safe reconciliation, discrepancy, receipt, retry, and non-critical pause controls without blocking safety information or altering clinical truth.
- ☐ Every listed objective test passes locally and at the required gate where applicable.
- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- ☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

1.4.4.12 Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	D-PAY-01: Alan - later processor, security and settlement mechanism
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	payment operations, authorization, observability
Higher-capability review	Mandatory before merge

1.5 Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain

Merge order inside this package: `P08-01` → `P08-02` → `P08-03` → `P08-04`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

1.6 Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Webhook replay/order; Ledger invariants; Milestone idempotency; Refund table; Node 0 credit floor; Reconciliation mismatch	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.

Test dimension	Method	Objective result
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

1.7 Risks and decisions

Founder outcomes, Alan's remaining technical ownership and external activation dependencies are recorded separately so a final policy is never mistaken for an active control.

ID	Status	Classification	Remaining owner	Affected / latest	Programme effect
D-COM-01	Founder decision final for beta; tax externally dependent	Luke decides commercial outcome	Alan - implementation	P06-01, P07-01, P08-01; Before activation of a Complete configuration	P06-01, P07-01 and P08-01 may proceed synthetically; live activation remains gated.
D-COM-02	Founder decision final for v1; live activation gated	Luke decides commercial outcome	Alan - implementation	P07-01, P08-01, P08-03; Before P07-01 activation data	Synthetic commercial configuration and ledgers may proceed; real charging remains gated.
D-COM-03	Externally dependent; founder policy inputs settled	External specialist / later evidence required	External legal/privacy/finance; Luke final business approval	P07-01, P07-02, P08-03; Before P07-02 user acceptance	Synthetic A2 preparation may proceed; live P07-02/P08 charging and B1 customer acceptance remain blocked.

ID	Status	Classification	Remaining owner	Affected / latest	Programme effect
D-COM-04	Founder decision final; implementation evidence pending	Luke decides commercial/product outcome	Alan - implementation	P07-02, P07-03, P07-04, P08-03, P09-04; Before P07-02	P07-02 through P09-04 may proceed under the settled semantics.
D-PAY-01	Founder constraints final; Alan later	Luke sets constraints; Alan decides later	Alan - later before P08-02	P08-02, P08-03, P08-04; Before P08-02 adapter	P08-01 ledger work may proceed; P08-02 and A2 payment completion wait for Alan's processor decision and evidence.
D-COM-05	Founder decision final for beta	Luke decides beta scope	None during beta	P07-02, P08-02; Before financed option is exposed	Financing no longer blocks P08-02; the feature is deliberately out of scope.
D-COM-06	Founder policy final; legal/accounting externally dependent	Luke decides policy; specialists translate	Alan - implementation; legal/accounting - wording	P08-03; Before automated pregnancy-stage refund logic	Synthetic P08-03 implementation may proceed; live policy activation remains gated.
D-COM-07	Founder policy final; legal/accounting externally dependent	Luke decides policy; specialists translate	Alan - implementation; legal/accounting - wording	P08-03; Before real billing	Synthetic P08-03 implementation may proceed; live refund/cancellation terms remain gated.
D-COM-08	Founder decision final; implementation evidence pending	Luke decides customer treatment	Alan - implementation	P08-04; Before P08-04	P08-04 may proceed under the settled policy.

1.7.1 D-COM-01 - Node 0 Complete price

Field	Recorded decision or requirement
Decision state	Final beta policy; external tax wording pending
Founder outcome	Node 0 Complete is a voluntary $495 USD one-time upgrade paid by the IP or authorized IP account holder, never the GC, collected before Complete features and fully credited against the later Phase 1/JMF without gating matching or safety support.

Field	Recorded decision or requirement
Founder constraints	Transparent one-time feature/credit wording; Versioned prospective changes only; No forced payment to identify, approach or reach agreement with a GC
What remains for Alan	Implement configurable price, payer ownership, durable payment/credit ledger and idempotent flows
Evidence required	Synthetic success, failure, retry, duplicate and credit tests; Tax wording before live activation
Approval rule	Luke approves future commercial-policy changes; tax treatment requires specialist review.
External dependencies	Tax treatment and customer-facing wording before live activation
Affected PRs / latest point	P06-01, P07-01, P08-01; Before activation of a Complete configuration
Programme effect	P06-01, P07-01 and P08-01 may proceed synthetically; live activation remains gated.
Independent recommendation	Keep no active production value until explicitly approved; test with synthetic configuration.

1.7.2 D-COM-02 - Ratify JMF price book and effective date

Field	Recorded decision or requirement
Decision state	Final v1 policy; external activation dependencies
Founder outcome	Configuration v1 is $8,900 USD: a $750 deposit followed by $1,630, $2,445, $1,630, $1,630 and $815 milestones. Pay-in-full is $7,830 total, leaving $7,080 after the deposit; the separate $495 Complete credit must not be double-counted.
Founder constraints	Versioned configuration; Prospective changes only; First live beta requires A2 plus approved terms and tax treatment
What remains for Alan	Implement versioned configuration, milestone and reconciliation behavior
Evidence required	Synthetic schedule, pay-in-full and no-double-credit reconciliation tests
Approval rule	Luke approves future price-book versions and first live activation.
External dependencies	Approved terms and tax treatment before live activation
Affected PRs / latest point	P07-01, P08-01, P08-03; Before P07-01 activation data
Programme effect	Synthetic commercial configuration and ledgers may proceed; real charging remains gated.
Independent recommendation	Use the document anchors but require explicit version activation rather than code literals.

1.7.3 D-COM-03 - Terms, privacy and refund language versions

Field	Recorded decision or requirement
Decision state	Externally dependent
Founder outcome	No live acceptance or charge may rely on the contradictory current terms draft. D-COM-06 and D-COM-07 supply the approved business policy for a new versioned terms package.
Founder constraints	Resolve refundable-until-onboarding versus non-refundable contradiction; No retrospective policy changes; Versioned terms/privacy/refund acceptance
What remains for Alan	Implement versioned acceptance and keep real charging disabled until an approved version is active
Evidence required	Approved paid-service terms; Approved privacy/consent disclosure; Approved tax/refund/accounting wording
Approval rule	Luke approves the final business policy and customer-facing version after specialist review.
External dependencies	Legal, privacy, tax and finance review before real-user acceptance or charging
Affected PRs / latest point	P07-01, P07-02, P08-03; Before P07-02 user acceptance
Programme effect	Synthetic A2 preparation may proceed; live P07-02/P08 charging and B1 customer acceptance remain blocked.
Independent recommendation	External legal/privacy/finance review before any real acceptance.

1.7.4 D-COM-04 - Enrollment funding and activation evidence

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	Use two-step commitment: settled $750 plus current accepted terms proves commercial enrollment, but payment alone never promotes. Exactly one enrollment-pending case follows explicit IP promotion and authenticated GC consent. The $1,630 activation tranche is earned only after defined IP/GC onboarding and screening coordination has begun; it never implies medical clearance.
Founder constraints	Redirects/webhooks never promote; Typed promotion event only; Source-labelled facts and explicit evidence checklist
What remains for Alan	Define typed events, activation checklist, idempotency and audit mechanics
Evidence required	Duplicate, replay, partial-failure, promotion and activation evidence tests

Field	Recorded decision or requirement
Approval rule	Luke approves any change to commercial enrollment or activation semantics.
External dependencies	None
Affected PRs / latest point	P07-02, P07-03, P07-04, P08-03, P09-04; Before P07-02
Programme effect	P07-02 through P09-04 may proceed under the settled semantics.
Independent recommendation	Funding confirmation is durable ledger evidence; promotion uses the sole typed authorization event; activation requires an explicit evidence checklist, actor and audit event.

1.7.5 D-PAY-01 - Payment processor and card/ACH scope

Field	Recorded decision or requirement
Decision state	Final founder constraints
Founder outcome	Offer card and US ACH, encourage but do not require ACH above $1,000, absorb ordinary beta fees without surcharging, and keep processor metadata free of GC, journey, clinical or document detail.
Founder constraints	Card cost ceiling near 3% plus $0.30; ACH cost ceiling near 1% capped at $10; Processor-neutral domain and ledger contract
What remains for Alan	Choose the processor/vendor, tokenization, webhook, settlement, return/refund and reconciliation mechanisms
Evidence required	Provider security and cost comparison; Synthetic settlement, return, refund, duplicate and reconciliation tests
Approval rule	Luke approval is required only if the proposed processor exceeds the cost/privacy constraints or changes the payment experience.
External dependencies	Payment/security/finance review before real money
Affected PRs / latest point	P08-02, P08-03, P08-04; Before P08-02 adapter
Programme effect	P08-01 ledger work may proceed; P08-02 and A2 payment completion wait for Alan's processor decision and evidence.
Independent recommendation	Choose only after domain/ledger contract; retain processor-neutral ports.

1.7.6 D-COM-05 - Financing path

Field	Recorded decision or requirement
Decision state	Final beta scope

Field	Recorded decision or requirement
Founder outcome	Financing is excluded and hidden from beta; Natalyx does not advertise, recommend or integrate a lender or treat lender status as funding evidence.
Founder constraints	Card and ACH remain unaffected; Reconsider only after demand evidence plus legal/privacy review
What remains for Alan	Keep financing paths disabled and absent from beta UI/configuration
Evidence required	Negative configuration/UI tests proving financing is unavailable
Approval rule	A future financing proposal requires a new founder decision.
External dependencies	None
Affected PRs / latest point	P07-02, P08-02; Before financed option is exposed
Programme effect	Financing no longer blocks P08-02; the feature is deliberately out of scope.
Independent recommendation	Do not block card/ACH MVP; keep optional and disabled.

1.7.7 D-COM-06 - Pregnancy-loss credit or retained-revenue policy

Field	Recorded decision or requirement
Decision state	Final founder policy; external wording pending
Founder outcome	After post-heartbeat loss, never charge the $815 birth tranche; retain the delivered $1,630 pregnancy-support tranche and grant an equal $1,630 future credit to the same IP household for 36 months, with compassionate manual extension and defined offboarding support.
Founder constraints	No automatic birth charge; Auditable credit expiry/extension; Compassionate support workflow
What remains for Alan	Implement credit ledger, expiry, extension and support workflow
Evidence required	Loss, credit, expiry, extension and duplicate tests; Approved legal/accounting treatment before live use
Approval rule	Luke approves final customer-facing policy after specialist review.
External dependencies	Legal and accounting review before live use
Affected PRs / latest point	P08-03; Before automated pregnancy-stage refund logic
Programme effect	Synthetic P08-03 implementation may proceed; live policy activation remains gated.
Independent recommendation	Manual/counsel-reviewed hook until policy is approved; never hard-code.

1.7.8 D-COM-07 - Cancellation and non-clearance policy sign-off

Field	Recorded decision or requirement
Decision state	Final founder policy; external wording pending
Founder outcome	Protect delivered work: the $750 is refundable until onboarding is delivered and then earned; activation follows D-COM-04. Non-clearance or pre-clearance GC withdrawal stops later charges. Unearned prepayment is refunded within ten business days, with refund or rollover choice and one replacement within 12 months without a second $750 deposit.
Founder constraints	A new activation charge may apply for genuinely new onboarding; No double charge or silent retained unearned value; Refund and rollover are auditable
What remains for Alan	Implement reconciliation, refund, rollover and replacement behavior
Evidence required	Withdrawal, non-clearance, refund, rollover, replacement and duplicate tests; Approved legal/accounting treatment before live use
Approval rule	Luke approves final customer-facing policy after specialist review.
External dependencies	Legal and accounting review before live use
Affected PRs / latest point	P08-03; Before real billing
Programme effect	Synthetic P08-03 implementation may proceed; live refund/cancellation terms remain gated.
Independent recommendation	Use document schedule as test anchor; require legal/finance approval for live policy.

1.7.9 D-COM-08 - Grace period and non-critical service pause

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	Use a humane 30-day grace period with reminders seven days before due, on the due date, and at 7, 14 and 30 days overdue; no beta late fee and human outreach before pause. Only future non-critical paid work may pause after 30 days.
Founder constraints	Never block safety content, existing documents/data, correction/export/deletion, consent/revocation, urgent support or already-required provider coordination
What remains for Alan	Implement reminder, escalation, grace and capability-specific pause states
Evidence required	Reminder cadence and protected-access tests
Approval rule	Luke approves changes to grace or protected access.
External dependencies	None
Affected PRs / latest point	P08-04; Before P08-04
Programme effect	P08-04 may proceed under the settled policy.

Field	Recorded decision or requirement
Independent recommendation	Default to continued safety and entitled-document access; pause only clearly non-critical future services.

Implementation risk	Owner	Treatment
Double charge/refund	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Processor lock-in	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Clinical timing harm	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Pass-through commingling	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Policy gaps	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.

1.8 Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
No direct source	-	Build from approved contracts/evidence	Package PRs	New implementation	N/A

Package source inventory:

- Business-model schedules and smoke paths; no current code is authoritative

1.9 Ownership recommendation

Luke is the proposed DRI and primary implementer for payment-domain work. Alan reviews and owns D-PAY-01's later processor/security mechanism; payment, finance, security and legal review remain required before live money movement.

PR	WP DRI	Primary implementer	Technical-decision owner	Reviewer	Final approver	Exception
P08-01	Luke	Luke (proposed)	None; implement the settled founder constraints and approved contracts	Alan	Luke	None
P08-02	Luke	Luke (proposed)	D-PAY-01: Alan - later processor, security and settlement mechanism	Alan	Luke	None
P08-03	Luke	Luke (proposed)	D-PAY-01: Alan - later processor, security and settlement mechanism	Alan	Luke	None
P08-04	Luke	Luke (proposed)	D-PAY-01: Alan - later processor, security and settlement mechanism	Alan	Luke	None

PR ownership inherits the proposed work-package split pending Alan's review. Named technical-decision ownership remains separate and never transfers implementation ownership across work packages; reciprocal and external review requirements remain intact.

1.10 Definition of done

- ☐ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and review record complete.
- ☐ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- ☐ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- ☐ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- ☐ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- ☐ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs

without team confirmation.

- ☐ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.